

Weiyong Xu

Postdoctoral Researcher in Cognitive Neuroscience

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🏛 Department of Psychology, University of Jyväskylä 🌐 Chinese

Research Interests

Topics: Memory reorganization, Sleep and Rest, Bodily Rhythms and Learning, Audiovisual Learning

Methods: MEG, EEG, Machine Learning, Calcium Imaging (One-photon), Behavioral Measures

Education

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|------------|--|------------------------|
| PhD | University of Jyväskylä , Psychology
<ul style="list-style-type: none"> Department: Department of Psychology Thesis: Brain activity changes related to learning of audiovisual associations in reading | Finland
2015 – 2020 |
| MS | Xi'an Jiaotong University , Biomedical Engineering
<ul style="list-style-type: none"> Department: Institute of Biomedical Engineering Thesis: Functional connectivity and complex network analysis of the aging brain | China
2013 – 2015 |
| BS | Xi'an Jiaotong University , Biomedical Engineering
<ul style="list-style-type: none"> Department: School of Life Science and Technology Thesis: EEG functional connectivity of vascular dementia patients in an oddball task | China
2009 – 2013 |

Work Experience

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|---|----------------------------------|
| Department of Psychology, University of Jyväskylä , Postdoctoral Researcher
Project: The reorganization of memory representations during rest and sleep
<ul style="list-style-type: none"> Study memory reorganization during rest and sleep using human neuroimaging methods (MEG/EEG), machine learning, and animal models (One-photon calcium imaging in mice). | Finland
Sept 2022 – Aug 2025 |
| Department of Psychology, University of Jyväskylä , Postdoctoral Researcher
Project: Bodily rhythms and learning
<ul style="list-style-type: none"> Investigated the interaction between physiological rhythms (such as heartbeat and breathing) and learning processes using behavioral and brain measures (M/EEG) in the Academy of Finland project "Optimizing Learning: Synchrony of the Brain and Body as a Tool?" (Dr. Miriam Nokia's group). Set up new calcium imaging systems and developed a data analysis pipeline for the Academy of Finland project "Dentate gyrus the gateway to memory?" (Dr. Markku Penttonen). | Finland
Sept 2020 – Aug 2022 |
| Department of Psychology, University of Jyväskylä , Doctoral Student
Project: Brain activity changes related to learning of audiovisual associations in reading
<ul style="list-style-type: none"> Doctoral Researcher (2019-09 2020-09) Early Stage Researcher (2015-09 2019-09), ChildBrain Project, Horizon2020 Marie Skłodowska-Curie Action (MSCA) Innovative Training Network (ITN) | Finland
Sept 2015 – Sept 2020 |

Research Visits

Visiting Researcher

Aalto University

- Department: Department of Neuroscience and Biomedical Engineering
- Host: Prof. Lauri Parkkonen

Finland
Feb 2025 – Mar 2025

Visiting Researcher

Tohoku University

- Host: Prof. Hyeonjeong Jeong

Japan
May 2025 – May 2025

Visiting Student

Donders Institute for Brain, Cognition and Behaviour

- Host: Prof. Robert Oostenveld
- Learn MEG data analysis methods

Netherlands
July 2016 – July 2016

Visiting Student

Aston Brain Center

- Host: Prof. Joel B. Talcott and Dr. Caroline Witton
- Learn child MEG and MRI measurements and data analysis

UK
Feb 2018 – Feb 2018

Funding

The reorganization of memory representations during rest and sleep

Academy of Finland Postdoctoral project

- Role: Principal Investigator
- Amount: 234,782 EUR

Sept 2022 – Aug 2025

Invited Talks

Neural correlates of retrospective memory confidence during face-name associative learning

26th Human Brain Science Seminar, Tohoku University

Sendai, Japan
2025

Oral Presentations

Neural correlates of retrospective memory confidence during face-name associative learning

The Learning Brain Across the Lifespan: Neural Resources, Imaging Indicators and Relevance for Cognitive Competence, Learning and Plasticity Meeting 2026

Ylläs, Finland
2026

From Novice to Expert: Different Levels of Learning in the Brain Revealed by MEG

Jyväskylä Brain & Mind Annual Seminar

Jyväskylä, Finland
Jan 2026

Neural correlates of retrospective memory confidence during face-name associative learning

International Conference on Cognitive Neuroscience (ICON 2025)

Porto, Portugal
2025

The reorganization of memory representations during rest and sleep

Neuroscience and Biomedical Engineering Seminar, Aalto University

Espoo, Finland
2025

Neural correlates of retrospective memory confidence during face-name associative learning

Staresina Lab Meeting

Online
2025

Differential brain processes of newly learned and overlearned audiovisual associations

Jyväskylä Brain & Mind Annual Seminar 2024

Jyväskylä, Finland
2024

Neural correlates of retrospective memory confidence during face-name associative learning MEG Nord 2023	Jyväskylä, Finland 2023
Brain responses to letters and speech sounds and their correlations with cognitive skills related to reading in children ChildBrain Conference 2018	Leuven, Belgium 2018
Audiovisual processing of Chinese characters elicits integration and congruency effects in MEG MEG Nord 2018	Stockholm, Sweden 2018

Poster Presentations

Differential brain processes of newly-learned and overlearned audiovisual associations Brains and Bodies in Social Interaction, Learning, and Wellbeing Conference 2024	Jyväskylä, Finland 2024
Differential brain processes of newly-learned and overlearned audiovisual associations FENS Forum 2024	Vienna, Austria 2024
Changes in brain activity during learning of grapheme-phoneme associations Society for the Neurobiology of Language (SNL 2019)	Helsinki, Finland 2019
Changes in brain activity during learning of grapheme-phoneme associations MEG Nord 2019	Jyväskylä, Finland 2019
Brain responses to letters and speech sounds and their correlations with cognitive skills related to reading in children Society for Neuroscience (SfN 2018)	San Diego, USA 2018
Brain responses to letters and speech sounds and their correlations with cognitive skills related to reading in children International Summer School "Life after PhD"	Jyväskylä, Finland 2018
Long-term learning effect on audio-visual processing of Chinese characters International Conference on Cognitive Science (ICCS 2017)	Taipei, Taiwan 2017
Development of audio-visual integration in children International Summer School "Understanding Learning in the Brain"	Jyväskylä, Finland 2017
Audio-visual processing of written characters and speech sounds MEG Nord 2017	Aarhus, Denmark 2017
Audio-visual processing of Chinese characters and speech sounds Biomag 2016 (20th International Conference on Biomagnetism)	Seoul, South Korea 2016

Workshops, Seminars, and Training

Cutting EEGX 2024 Workshop	Online 2024
Brain & Mind Symposium 2023 Workshop	Espoo, Finland 2023
eSleep Europe 2023 Workshop	Online 2023

Brain and Language 2021: Virtual Conference Methods Workshop Workshop	Jyväskylä, Finland 2021
6th Annual Research Seminar of CIBR Workshop	Jyväskylä, Finland 2018
ChildBrain 6th Workshop, Donders Institute Workshop	Nijmegen, Netherlands 2018
5th Annual Research Seminar of CIBR Workshop	Jyväskylä, Finland 2017
ChildBrain 5th Workshop, Aston University Workshop	Birmingham, UK 2017
Nordic & Baltic Doctoral Programmes in Psychology Joint Symposium Workshop	Tartu, Estonia 2017
ChildBrain 4th Workshop, Donders Institute Workshop	Nijmegen, Netherlands 2017
4th Annual Research Seminar of CIBR Workshop	Jyväskylä, Finland 2016
ChildBrain 3rd Workshop Workshop	Leuven, Belgium 2016
Toolkit of Cognitive Neuroscience: Advanced data analysis and source modelling of EEG/MEG Workshop	Nijmegen, Netherlands 2016
Winter School on Neural Mechanisms of Learning Workshop	Jyväskylä, Finland 2016
Elekta Neuromag Triux MEG System Start Program, CIBR Workshop	Jyväskylä, Finland 2016
ForLearning International Symposium on Learning and Learning Difficulties Workshop	Jyväskylä, Finland 2015
ChildBrain 1st Workshop (Kick-off) Workshop	Münster, Germany 2015
MEG Driving License Workshop	University of Jyväskylä, Finland 2015

Teaching

PSYS2102 Master's thesis ending seminar instructor • University of Jyväskylä	2023
CNSJ1202 Designing cognitive neuroscience research instructor • University of Jyväskylä	2022

CIBA120 EEG data analysis Demo using MNE Python demo instructor • University of Jyväskylä	2022
CIPA113 Bilingual Development essays grading • University of Jyväskylä	2021
CIPA112 Psychology of Language essays grading • University of Jyväskylä	2021
CIPA213 Developmental Language Related Disorders instructor • University of Jyväskylä	2019
CIPA100 Psychology Research at JYU essays grading • University of Jyväskylä	2016 – 2019
PSYS123 Neuropsychology 3B MEG demonstrations instructor • University of Jyväskylä	2017
Introduction to scripting experiments with Presentation instructor • University of Jyväskylä	2016

Awards

Winner of the best poster competition Annual Research Seminar of Centre for Interdisciplinary Brain Research (CIBR)	2017
Siyuan Student Scholarship Xi'an Jiaotong University	2010 – 2013

Service and Outreach

Membership: Brain Research Society of Finland (BRSF), Federation of European Neuroscience Societies (FENS), International Brain Research Organisation (IBRO), Society for the Neurobiology of Language (SNL), Society for Neuroscience (SfN)

Ad Hoc Reviewer: NeuroImage, Frontiers in Psychology, Journal of Neuroscience, Child Development, Journal of Neurolinguistics, Computer Methods and Programs in Biomedicine, Acta Psychologica, npj Science of Learning, Journal of Open Source Software, Neuropsychologia

Supervision

Master's: Master theses (1): Jukka Lopmeri (ongoing)

Bachelor's: Bachelor's theses (3)

Doctoral: Co-supervisor for 3 doctoral students: Jiaqi Zheng, Aino Sorsa, Suvi Lehtonen.

Skills

Languages: Chinese (Mother tongue), English (Proficient)

Programming: Python, MATLAB, R

Neuroimaging: MEG analysis (MNE-Python, FieldTrip), EEG analysis, Calcium imaging analysis, Machine learning for neuroimaging

Tools: Git/GitHub, LaTeX, Linux/Unix

Scientific and Societal Impact

- Outreach: Organize activities for Researcher's night events at JYU (2016, 2017, 2020, 2021, 2022, 2024, 2025)
- Tours: MEG lab tours for Finnish high school students, Bachelor and Master level students (2017)
- Tours: MEG lab tours for the Finnish Psychology Students' Association (2017)
- Events: Brain awareness week event "Developing Brain" at JYU (Poster presentation) (2016)

Publications

- Assay of melamine in milk products with a pH-mediated stacking technique in capillary electrophoresis** Mar 2014
Kong, Y., Yuan, J., Wang, Z., Hou, Z., Yu, J., Jin, H., Xie, M., Chang, N., **Xu, W.**, Zhao, Y., Tang, Y.
[10.1002/jssc.201300603](https://doi.org/10.1002/jssc.201300603) (Journal of Separation Science, 37(6), 717-724)
- Brain Responses to Letters and Speech Sounds and Their Correlations With Cognitive Skills Related to Reading in Children** Aug 2018
Xu, W., Kolozsvári, O. B., Monto, S. P., Hämäläinen, J. A.
[10.3389/fnhum.2018.00304](https://doi.org/10.3389/fnhum.2018.00304) (Frontiers in Human Neuroscience, 12, Article 304)
- Aging-related changes of EEG synchronization during a visual working memory task** Dec 2018
Teng, C., Cheng, Y., Wang, C., Ren, Y., **Xu, W.**, Xu, J.
[10.1007/s11571-018-9500-6](https://doi.org/10.1007/s11571-018-9500-6) (Cognitive Neurodynamics, 12(6), 561-568)
- Audiovisual Processing of Chinese Characters Elicits Suppression and Congruency Effects in MEG** Feb 2019
Xu, W., Kolozsvári, O. B., Oostenveld, R., Leppänen, P. H. T., Hämäläinen, J. A.
[10.3389/fnhum.2019.00018](https://doi.org/10.3389/fnhum.2019.00018) (Frontiers in Human Neuroscience, 13, Article 18)
- Top-Down Predictions of Familiarity and Congruency in Audio-Visual Speech Perception at Neural Level** July 2019
Kolozsvári, O. B., **Xu, W.**, Leppänen, P. H. T., Hämäläinen, J. A.
[10.3389/fnhum.2019.00243](https://doi.org/10.3389/fnhum.2019.00243) (Frontiers in Human Neuroscience, 13, Article 243)
- Context-dependent minimisation of prediction errors involves temporal-frontal activation** Feb 2020
Hsu, Y. F., **Xu, W.**, Parviainen, T., Hämäläinen, J. A.
[10.1016/j.neuroimage.2019.116355](https://doi.org/10.1016/j.neuroimage.2019.116355) (NeuroImage, 207, Article 116355)
- Rapid changes in brain activity during learning of grapheme-phoneme associations in adults** Oct 2020
Xu, W., Kolozsvári, O. B., Oostenveld, R., Hämäläinen, J. A.
[10.1016/j.neuroimage.2020.117058](https://doi.org/10.1016/j.neuroimage.2020.117058) (NeuroImage, 220, Article 117058)
- Neural Responses to Musical Rhythm in Chinese Children With Reading Difficulties** June 2020
Chiang, C. H., Hämäläinen, J., **Xu, W.**, Wang, H. L.
[10.3389/fpsyg.2020.01013](https://doi.org/10.3389/fpsyg.2020.01013) (Frontiers in Psychology, 11, Article 1013)
- Brain activity changes related to learning of audiovisual associations in reading** 2020
Xu, W.
urn.fi/URN:ISBN:978-951-39-8220-1 (JYU Dissertations 249, University of Jyväskylä)
- Coherence Between Brain Activation and Speech Envelope at Word and Sentence Levels Showed Age-Related Differences in Low Frequency Bands** May 2021
Kolozsvári, O. B., **Xu, W.**, Gerike, G., Parviainen, T., Nieminen, L., Noiray, A., Hämäläinen, J. A.
[10.1162/nol_a_00033](https://doi.org/10.1162/nol_a_00033) (Neurobiology of Language, 2(2), 226-253)

Cardiac cycle and respiration phase affect responses to the conditioned stimulus in young adults trained in trace eyeblink conditioning Waselius, T., Xu, W. , Sparre, J. I., Penttonen, M., Nokia, M. S. 10.1152/jn.00298.2021 (Journal of Neurophysiology, 127(3), 767-775)	Mar 2022
Cardiorespiratory rhythm-contingent trace eyeblink conditioning in elderly adults Santhana Gopalan, P. R., Xu, W. , Waselius, T., Wikgren, J., Penttonen, M., Nokia, M. S. 10.1152/jn.00356.2023 (Journal of Neurophysiology, 131(5), 797-806)	May 2024
Neural correlates of retrospective memory confidence during face-name associative learning Xu, W. , Li, X., Parviainen, T., Nokia, M. 10.1093/cercor/bhae194 (Cerebral Cortex, 34(5), Article bhae194)	May 2024
The cardiac cycle modulates learning-related interoception Nokia, M. S., Xu, W. , Wikgren, J. 10.1016/j.tics.2024.05.007 (Trends in Cognitive Sciences, 28(8), 691-692)	Aug 2024
Optimizing learning - synchrony of the brain and body as a tool? Nokia, M., Xu, W. 10.17011/jyx/dataset/98410 (JYX, University of Jyväskylä)	Nov 2024
Neural signatures of automatic letter-speech sound integration in literate adults Xu, W. , Li, X., Kolozsvari, O., Sorsa, A., Nokia, M., Hämäläinen, J. 10.1162/imag.a.1021 (Imaging Neuroscience, 3, Article IMAG.a.1021)	Nov 2025
Dynamic changes in electrical brain activity during passive learning of foreign speech sound discrimination Li, Q., Kurkela, J. L. O., Lohvansuu, K., Hämäläinen, J. A., Li, X., Xu, W. , Astikainen, P. 10.1093/cercor/bhaf181 (Cerebral Cortex, 35(7), Article bhaf181)	July 2025
Two-Day Artificial Orthography Learning MEG Dataset: From Grapheme to Meaning Xu, W. , Kolozsvari, O., Hämäläinen, J., Oostenveld, R. 10.17011/jyx/dataset/106229 (JYX, University of Jyväskylä)	Oct 2025
Development of brain responses to speech sounds and associations with reading and reading related skills in adults with childhood dyslexia Sorsa, A., Tretow, A., Xu, W. , Gerike, G., Lohvansuu, K., Hämäläinen, J. A., Leppänen, P. H. T. 10.1016/j.ijpsycho.2026.113339 (International Journal of Psychophysiology, 222, Article 113339)	Apr 2026
Magnetoencephalography Source Connectivity Pipeline Reveals Long-Range Connectivity in Audio-Visual Integration Bornfleth, H., Cho, J. H., Rusiniak, M., Kornweibel, O., Xu, W. , Hämäläinen, J., Spangler, R. 10.1111/ejn.70470 (European Journal of Neuroscience, 63(7), Article e70470)	Apr 2026